CC: Sports: Sports Training and Testing

Module 1: Fundamentals of Sports Training

Q1. Define sports training. Explain its features and aims. [5 Marks]

Answer:

Sports training is the systematic, planned and progressive process of preparing an athlete for the highest possible performance in a chosen sport through physical exercise, skill practice, tactical, psychological and theoretical preparation. Hardayal Singh describes it as a pedagogical process based on scientific principles; Bompa calls it a long-term, progressively and individually graded activity.

- **Systematic and scientific:** organised in a logical order and based on physiology, biomechanics, psychology and nutrition (e.g. heart-rate zones for endurance).
- **Progressive and individualised:** load rises gradually and is adapted to each athlete's age and ability.
- **Long-term, goal-directed and all-round:** it continues over years towards a specific target and develops fitness, skill, tactics and mind together.
- **Pedagogical:** the athlete learns why training, recovery and nutrition matter.
- **Training vs exercise:** exercise is any planned activity for fitness (a daily walk); training is exercise organised over time towards a performance goal, with progression and monitoring.
- **Aims:** improve performance; develop fitness, technique and tactics; build psychological qualities; prevent injury; build knowledge; promote health; and develop discipline, teamwork and fair play.
- **Importance:** tiny winning margins (Noah Lyles won the Paris 2024 100 m by 0.005 s); health benefits in line with WHO guidelines; injury prevention (FIFA 11+); national success through TOPS and Khelo India.

Example: Neeraj Chopra's Olympic gold (Tokyo 2020) came from years of systematic strength, technique and periodised training.

Q2. Explain the components of sports training. [5 Marks]

Answer:

Bompa groups sports training into five areas of preparation, all of which must develop together for peak performance.

- **Physical preparation:** develops motor abilities — strength, speed, endurance, flexibility and coordination. It includes general physical preparation (GPP, an overall fitness base) and specific physical preparation (SPP, sport-specific qualities such as repeated sprint ability for footballers).
- **Fitness components:** health-related (cardiorespiratory endurance, muscular strength and endurance, flexibility, body composition) and skill-related (speed, agility, power, balance, coordination, reaction time).
- **Technical preparation:** learning and refining the most efficient way to perform a movement, through the cognitive, associative and autonomous stages (Fitts and Posner), using video and feedback. Example: PV Sindhu refining her strokes under high heart rate.
- **Tactical preparation:** plans and decisions to outwit opponents — field placements, pressing in football, pacing in a 1500 m race — trained through game simulations and match analysis.
- **Psychological preparation:** motivation, goal-setting, concentration, confidence and emotional control under pressure.
- **Theoretical preparation:** knowledge of the sport, rules, the body, nutrition and recovery, so the athlete understands and commits to training.

A cricketer, for instance, needs fitness, batting technique, match strategy and concentration all at once.

Q3. Explain the FITT principle and the principle of specificity with examples. [5 Marks]

Answer:

Principles of training are scientific guidelines on how the body responds to exercise. They produce consistent improvement, prevent injury and overtraining, and help individualise programmes.

- **FITT:** a framework for prescribing exercise — Frequency (how often), Intensity (how hard: % HRmax, RPE, % 1RM), Time (how long, or sets and reps) and Type (running, cycling, weights, yoga). ACSM extends it to FITT-VP by adding Volume and Progression.
- **FITT example:** a sedentary student may start with 3 days a week of brisk walking or walk-jog at 55–65% HRmax for 20 minutes, progressing to 4–5 days at 65–80% HRmax for 40 minutes by weeks 9–12.
- **Beating plateaus:** change one FITT variable at a time, e.g. add hills while keeping frequency and time the same.
- **Specificity (SAID principle):** Specific Adaptations to Imposed Demands — the body adapts to the exact demands placed on it.
- **Specific to:** energy system (aerobic for a marathoner, ATP-PC for a sprinter), muscle groups, movement pattern, speed, range of motion, contraction type and environment (heat acclimatisation).
- **Transfer of training:** exercises similar to the sport in movement, force and speed transfer best (e.g. rotational medicine-ball throws for a javelin thrower).
- **Balance:** specificity does not mean only doing the sport — general base training comes first, especially for young athletes.

Examples: kabaddi players train explosive bursts and grip; Eliud Kipchoge builds huge aerobic capacity through high running volume.

Q4. Explain the principles of overload, progression and reversibility. [5 Marks]

Answer:

These principles rest on stress, adaptation and recovery. Training disturbs homeostasis; during recovery the body adapts (supercompensation). Selye's General Adaptation Syndrome describes alarm, resistance and exhaustion stages, and the fitness–fatigue model states that Performance = Fitness − Fatigue.

- **Overload:** to improve, the body must face a load greater than it is used to. It can be applied by increasing intensity, duration or frequency, reducing rest, or adding complexity.
- **Milo of Croton:** the legendary Greek wrestler who carried a calf daily as it grew into a bull — the classic illustration of progressive overload.
- **Optimal overload:** too much without recovery leads to overreaching and possibly overtraining syndrome.
- **Progression:** overload must increase gradually. Guides include the roughly 10% weekly rule for runners, the NSCA '2-for-2 rule' in strength training, and Bompa's step loading (3 weeks up, 1 lighter week).
- **Too slow vs too fast:** too slow causes plateaus and boredom; too fast causes fatigue, injury and overtraining.
- **Reversibility ('use it or lose it'):** gains are lost when training stops. Aerobic fitness drops within 2–4 weeks; strength lasts longer, while power and speed fall faster; flexibility also declines.
- **Maintenance:** fitness can be kept with reduced frequency and volume if intensity is maintained — useful during exams, injury (e.g. cycling while a leg heals), COVID-19 lockdowns or the off-season.

Together, these principles ensure steady, safe improvement and the retention of fitness gains.

Q5. Compare continuous training and interval training. [5 Marks]

Answer:

Choice of method depends on the energy system a sport uses — ATP-PC (up to about 10 s), anaerobic glycolysis (10 s to 2 min) or aerobic (beyond 2 min).

- **Continuous training — meaning:** exercising without rest for a long period at low to moderate intensity to build aerobic endurance.
- **Types:** slow continuous/long slow distance (60–70% HRmax), tempo runs (80–90% HRmax near lactate threshold), fartlek ('speed play', developed by Gösta Holmér in Sweden) and cross-country running.
- **Effects and use:** larger stroke volume, lower resting heart rate, more mitochondria and capillaries, better fat use and higher VO_2max. Simple and good for beginners, but can be monotonous and does not develop speed. Suits marathons, cycling and swimming.
- **Interval training — meaning:** alternating high-intensity work with rest or light recovery, allowing more high-intensity work in total. Developed by Gerschler and Reindell (Freiburg method); Emil Zátopek was famous for it.
- **Variables:** work distance or duration, intensity, recovery duration and type, repetitions, sets and work:rest ratio.
- **Types:** extensive intervals (aerobic), intensive intervals (anaerobic), repetition sprints (full recovery), HIIT, and the Tabata protocol (20 s work, 10 s rest × 8).
- **Effects and use:** raises VO_2max, lactate tolerance and repeated-sprint ability, and is time-efficient; but it is demanding and riskier without a base. Suits middle-distance running and team and racket sports.

Continuous training builds the aerobic base; interval training builds on it for speed endurance.

Q6. Explain circuit training and flexibility training. [5 Marks]

Answer:

Circuit training develops general fitness and muscular endurance, while flexibility training improves the range of motion at joints.

- **Circuit training:** a series of 6–12 exercise stations performed one after another, by time (e.g. 30 s) or repetitions, with short rest and 2–4 rounds.
- **Organising a circuit:** alternate muscle groups (legs, upper body, core, cardio) so one recovers while another works; progress by adding time, reps or rounds, or cutting rest. Types include general fitness, strength, aerobic, sport-specific and bodyweight circuits.
- **Merits and limits:** develops several components at once, suits large groups and needs little equipment; but it is not ideal for maximal strength or speed, and technique may suffer with fatigue.
- **Flexibility training:** flexibility is the range of motion at a joint — static (held still) or dynamic (during movement).
- **Methods:** static stretching (hold 15–60 s, after training); dynamic stretching (controlled movements, in the warm-up); ballistic (bouncing, generally not recommended); PNF (contract–relax with a partner, large gains); and yoga or mobility work.
- **Evidence:** long static stretches (over about 60 s) right before explosive activity can reduce power, so use a dynamic RAMP warm-up (Raise, Activate, Mobilise, Potentiate) before, and static or PNF stretching after.
- **Benefits:** better range for skills (hurdlers, gymnasts, swimmers), posture, relaxation and reduced muscle tension.

Circuits are widely used in schools, police and military training; flexibility work is essential in gymnastics, martial arts and diving.

Q7. Describe the steps in designing a training plan and the structure of a training session. [5 Marks]

Answer:

A training plan is a written schedule that specifies what exercises to do, how much, how hard, how often and in what order over a period, to achieve specific goals.

- **1. Screening:** PAR-Q+, medical history and informed consent.
- **2. Needs analysis:** of the sport (movements, energy systems, common injuries) and of the athlete (training age, test results, strengths and weaknesses, injuries, lifestyle).
- **3. SMART goals:** e.g. 'Improve my Cooper distance from 2,200 m to 2,450 m in 12 weeks'. Distinguish outcome, performance and process goals.
- **4–5. Methods and variables:** select methods and set FITT-VP, rest intervals and exercise order (power, then multi-joint, then single-joint).
- **6. Periodisation:** macrocycle (year), mesocycle (2–6 weeks), microcycle (week); phases are preparatory (general and specific), competitive (with taper) and transition.
- **7–9. Session design, monitoring and re-testing:** keep a training diary, track fatigue, and re-test every 4–8 weeks to adjust the plan.
- **Session structure:** introduction (2–5 min); warm-up (10–20 min, RAMP); main part (skill and speed when fresh, then strength, then endurance); cool-down (5–15 min); and review (feedback, log RPE).

A good plan turns training principles and methods into organised daily practice.

Q8. What is training load? Explain how training intensity is measured. [5 Marks]

Answer:

Training load is the total stress placed on an athlete by training and competition. It combines volume, intensity and frequency, and is affected by density and complexity.

- **External vs internal load:** external load is the work done (km run, kg lifted, balls bowled); internal load is the body's response (heart rate, RPE, lactate). Two players doing the same 6 × 400 m session may have very different internal loads.
- **Session RPE:** session load = RPE (0–10) × duration in minutes; e.g. 7 × 60 = 420 AU. The acute:chronic workload ratio warns against sudden spikes in load.
- **Overtraining:** functional overreaching (days) and non-functional overreaching (weeks) can lead to overtraining syndrome (months). Warning signs include fatigue, falling performance, raised resting heart rate, poor sleep and frequent illness.
- **Intensity and volume:** intensity is how hard the athlete works; the two are inversely related.
- **Heart rate:** HRmax ≈ 220 − age (or Tanaka: 208 − 0.7 × age). Karvonen method: Target HR = HRrest + (HRmax − HRrest) × %. Example: 70 + (130 × 0.70) = 161 bpm.
- **RPE and talk test:** Borg 6–20 or CR-10 scales; at moderate intensity you can talk but not sing.
- **% 1RM:** strength ≥85% for ≤6 reps; hypertrophy 67–85% for 6–12 reps; endurance ≤67% for ≥12 reps. Speed, pace and lab markers such as lactate are also used.

Monitoring load and intensity helps coaches balance training stress with recovery.

Q9. Explain the principle of individual differences in sports training. [5 Marks]

Answer:

The principle of individuality states that no two people respond in exactly the same way to the same training, so programmes must be adapted to the individual.

- **Heredity:** genes influence VO_2max, muscle-fibre type and trainability. The HERITAGE Family Study found 'high' and 'low responders' to the same 20-week programme.

- **Age:** chronological and biological age differ — two 14-year-olds may be years apart in maturity; older adults recover more slowly.
- **Sex:** women generally have lower absolute strength but similar relative trainability; menstrual-cycle symptoms may call for individual adjustment.
- **Training age and initial fitness:** beginners and low-fitness individuals improve quickly; advanced athletes improve slowly and need more specific, varied stimuli.
- **Body type and health history:** heavier athletes may need lower-impact progression; a past ankle sprain calls for balance training.
- **Recovery capacity and lifestyle:** sleep, nutrition, exam stress, travel and climate (e.g. Indian summer heat) all affect adaptation.
- **Psychological factors:** motivation, confidence and anxiety — some athletes thrive under pressure, others need encouragement.
- **Application:** test athletes, set individual loads and zones, monitor responses, and adjust the plan.

Recognising individual differences makes training safer and more effective for every athlete.

Module 2: Tests and Measurement in Sports

Q10. Distinguish between test, measurement and evaluation, and explain the importance of tests and measurement in sports. [5 Marks]

Answer:

Tests and measurement turn impressions ('she looks fitter') into objective information ('her Cooper distance improved from 2,100 m to 2,380 m').

- **Test:** an instrument or procedure used to obtain information, e.g. the push-up test.
- **Measurement:** collecting data using a test, usually as a number, e.g. 'Riya completed 22 modified push-ups'.
- **Evaluation:** interpreting measurements to make a judgement, e.g. '22 is very good for her age'. The relationship is Test → Measurement → Evaluation.
- **Assessment:** a broader process combining tests, observation and records.
- **Types:** objective vs subjective; norm-referenced vs criterion-referenced; formative vs summative; field vs laboratory tests.
- **Importance:** assessing current status; diagnosing strengths and weaknesses; classifying students into groups; selecting athletes; motivating; monitoring progress; evaluating programmes; predicting performance; guiding training; and supporting research and grading.

Without measurement, coaches cannot know whether training is working or whom to select.

Q11. Explain validity, reliability and objectivity as characteristics of a good test. [5 Marks]

Answer:

A test is useful only if its results can be trusted. Validity, reliability and objectivity are the three main criteria of scientific authenticity.

- **Validity:** the degree to which a test measures what it is meant to measure. A 100 m sprint is invalid for aerobic endurance.
- **Types of validity:** face, content, criterion-related (concurrent, e.g. Cooper test vs lab VO_2max with $r \approx 0.90$; predictive) and construct validity.
- **Reliability:** consistency of scores on repeated testing. Errors come from the participant, tester, instrument, environment and procedure. It is estimated by test–retest, parallel forms, split-half, internal consistency or ICC; 0.90 and above is excellent.

- **Objectivity:** the degree to which different testers get the same scores (inter-tester reliability). It is improved by clear scoring rules, trained testers, multiple judges and calibrated equipment.
- **Relationship:** objectivity contributes to reliability, and reliability is necessary but not sufficient for validity — a test can be reliable without being valid, but not valid without being reliable.
- **Other qualities:** norms, practicality, economy, discrimination, standardised instructions, educational value, safety and appropriate difficulty.

A good test must be valid, reliable and objective at the same time.

Q12. Describe the Cooper 12-minute run/walk test. [5 Marks]

Answer:

Developed by Dr Kenneth H. Cooper in 1968 for the US Air Force, the Cooper test is a simple field test of cardiorespiratory endurance that estimates VO_2max.

- **Equipment:** a measured track (ideally 400 m), stopwatch, whistle, lap counters, cones, water and first aid.
- **Procedure:** warm up; on 'Go', cover as much distance as possible in 12 minutes, running or walking; time is announced at intervals; at the whistle, stop and stand still; record laps plus the partial lap; cool down.
- **VO_2max formula:** (distance in metres − 504.9) ÷ 44.73. For 2,400 m this gives about 42.4 ml/kg/min.
- **Norms (ages 20–29):** excellent is above 2,800 m for men and 2,700 m for women; poor is below 1,600 m and 1,500 m respectively.
- **Validity and reliability:** good concurrent validity with lab VO_2max; good reliability when conditions are standardised; high objectivity.
- **Factors affecting results:** motivation, pacing, weather (test early in Indian summers), surface, body mass and group running.
- **Merits and limits:** cheap and good for large groups, but it is a maximal test that needs health screening and good pacing. Walking tests like the Rockport test may suit unfit people.

Variations include the Cooper swim and cycle tests, the 2.4 km run, and the Yo-Yo or beep test for team sports.

Q13. Describe the sit-and-reach test and the push-up test. [5 Marks]

Answer:

These are two widely used field tests: the sit-and-reach test for flexibility and the push-up test for upper-body muscular endurance.

- **Sit-and-reach — purpose and origin:** measures hamstring and lower-back flexibility; introduced by Wells and Dillon (1952).
- **Sit-and-reach — procedure:** light warm-up; sit without shoes with legs straight and feet against the box; place hands one on top of the other; reach forward slowly without bouncing; hold 1–2 s; record to the nearest 0.5 cm; take the best of 2–3 trials. Use the norm table that matches the box.
- **Sit-and-reach — quality:** high reliability; moderate validity for hamstrings but lower for the back; limb length affects scores. Variations include the back-saver, modified, V-sit and chair versions.
- **Push-up — purpose:** measures muscular endurance of the chest, shoulders and triceps.
- **Push-up — procedure:** men use the standard position, women the modified knee position; lower the chest to about 10 cm (a fist) from the floor and push up to full extension; do as many correct reps as possible without rest; count only correct reps.
- **Push-up — norms and variations:** for ages 20–29, excellent is 36+ for men and 30+ for women. Variations include the timed 1-minute, cadence and 90° push-up tests.
- **Push-up — quality:** good face validity; objectivity depends on clear criteria for a correct rep, improved with a depth block.

Both tests are simple, low-cost and practical for testing large groups.

Q14. Explain the measurement of height and weight, and the calculation and limitations of BMI. [5 Marks]

Answer:

Anthropometry is the measurement of body size, shape and proportions. It is used for talent identification, weight categories, growth monitoring and health screening.

- **Height:** measured with a stadiometer; no shoes; heels, buttocks, shoulder blades and head against the scale; head in the Frankfort plane; take a deep breath; record to 0.1 cm. People are about 1 cm taller in the morning.
- **Weight:** measured on a calibrated scale on a hard floor, in light clothing, to 0.1 kg — ideally in the morning. Pre- and post-training weight estimates fluid loss (1 kg ≈ 1 litre).
- **BMI formula:** weight (kg) ÷ height (m)^2, proposed by Quetelet. Example: 68 ÷ 1.70^2 = 23.5.
- **WHO classification:** below 18.5 underweight; 18.5–24.9 normal; 25–29.9 overweight; 30 and above obese.
- **Indian cut-offs (2009):** normal 18.0–22.9, overweight 23.0–24.9, obese 25 and above, because Asian Indians carry more body fat at lower BMI. A BMI of 23.5 is therefore overweight by Indian criteria.
- **Children:** use age- and sex-specific BMI-for-age charts, not adult cut-offs.
- **Limitations:** BMI does not separate fat from muscle (a muscular wrestler may read as 'obese'), ignores fat distribution, and varies by ethnicity, age and sex. Waist circumference (risk above 90 cm for men, 80 cm for women) and waist-to-hip or waist-to-height ratios complement it.

BMI is a quick screening tool, but it should be interpreted with other measures.

Q15. How is fitness testing applied to improve performance and plan training? [5 Marks]

Answer:

Testing is useful only if results lead to action. It forms a continuous cycle: test → plan → train → re-test → adjust.

- **Steps:** select tests suited to the sport (needs analysis); do baseline testing; interpret results against norms, criteria and the athlete's own past scores; identify strengths and weaknesses; set SMART goals; design the programme; monitor; re-test every 4–12 weeks; and adjust.
- **Statistical tools:** percentiles; z-score = (score − mean) ÷ SD; T-score = 50 + 10z, which allows comparison across tests; and % change = (post − pre) ÷ pre × 100. Changes smaller than measurement error may not be real.
- **Fitness profile:** a radar or bar chart that shows strengths and weaknesses at a glance.
- **Linking results to training:** a low Cooper score calls for continuous training, then tempo runs and intervals; a low sit-and-reach score calls for static and PNF stretching; low push-ups call for circuits and push-up progressions; high BMI with a large waist calls for aerobic and resistance training plus nutrition guidance; declining scores may signal overtraining.
- **Setting intensities:** 2,400 m in 12 minutes means 200 m/min — easy runs at 65–75% of this speed, tempo at 85–90%, intervals at or slightly above it. Heart-rate zones can be set from HRmax or the Karvonen method.

Applied properly, fitness testing makes training individual, measurable and goal-directed.